

SmartFinance: A Gamified Cross-Platform Mobile App for Personal Budgeting & Investment Tracking for Malaysian Young Adults

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A project report submitted to the
Faculty of Computing and Information Technology
in partial fulfillment of the requirement for the
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Declaration

The project submitted herewith is a result of my own efforts in totality and in every aspect of the project works. All information that has been obtained from other sources had been fully acknowledged. I understand that any plagiarism, cheating or collusion or any sorts constitutes a breach of TAR University rules and regulations and would be subjected to disciplinary actions.



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Abstract

Financial literacy among Malaysian young adults remains critically low, with many lacking adequate financial knowledge essential for long-term stability. This project addresses the gap in culturally appropriate, engaging financial management tools by developing SmartFinance, a gamified cross-platform mobile application specifically designed for Malaysian young adults aged 18-30. The application combines practical financial management with behavioral design principles to encourage sustainable financial habits through user-friendly expense tracking, budget management, and basic investment portfolio monitoring.

The project employs Agile development methodology with user-centered design integration, utilizing Flutter for cross-platform mobile development, Python Flask for backend services, and MySQL for secure data management. The solution incorporates gamification elements including achievement badges, habit streaks, and progress celebrations to address psychological barriers that prevent consistent financial tracking among young adults. Core functionality includes intelligent expense categorization, visual budget monitoring, spending pattern analysis, and manual investment tracking without third-party banking integration to ensure data privacy and accessibility.

User feedback is gathered through Google Forms surveys distributed to university students and young professionals, with usability testing conducted among peers and fellow students to validate the user experience design and feature effectiveness. The application targets university students managing RM500-1500 monthly allowances and entry-level professionals earning RM2000-4000 monthly, addressing their specific financial literacy development needs.

The project contributes to Malaysian financial literacy improvement by providing culturally sensitive financial tools that understand local banking habits, cash-based transactions, and conservative investment approaches. Expected outcomes include improved user financial consistency through tracked behavior patterns, enhanced financial literacy through guided experiences, and sustainable habit formation through motivational design elements. The solution demonstrates practical application of Human-Computer Interaction principles and behavioral psychology in financial technology, serving as a reference framework for future culturally appropriate fintech solutions in Southeast Asia while addressing the critical need for accessible financial education tools among Malaysian emerging adults.

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Chapter 1

Introduction

1 Introduction

This chapter presents an overview of the SmartFinance project, a gamified cross-platform mobile application designed to address financial literacy challenges among Malaysian young adults. The chapter outlines the project objectives, background context, proposed solution, project scope, and organizational structure necessary to deliver a comprehensive personal budgeting and investment tracking solution.

1.1 Project Background

Young adults in Malaysia face considerable challenges when it comes to managing personal finances effectively. Despite widespread access to smartphones and digital technology, many struggle to develop consistent financial management habits. This disconnect between available technology and actual usage suggests that current financial management tools fail to address the specific needs and behavioral patterns of this demographic.

Sustainable Development Goals (SDG) Alignment: This project directly supports SDG 4 (Quality Education) by promoting financial literacy education through accessible digital tools. The application also contributes to SDG 8 (Decent Work and Economic Growth) by empowering young adults with financial management capabilities essential for economic participation and long-term financial stability. By providing educational financial tools, the project addresses the critical gap in practical financial education that traditional academic curricula often overlook.

1.1.1 Current Financial Management Challenges

The difficulty young adults face in maintaining consistent financial tracking habits stems from a complex interplay of psychological and practical factors. Behavioral economics research has shown that cognitive biases such as present bias — where immediate gratification is valued over future benefits — and loss aversion significantly undermine sustained financial management efforts (Kahneman & Tversky, 1979). Traditional financial tools often overlook these behavioral principles, focusing instead on feature-rich interfaces that can overwhelm users rather than motivate them.

Existing financial management applications present several limitations for Malaysian young adults. International solutions like Mint and You Need a Budget (YNAB) are designed around Western banking systems, featuring automatic bank integration and credit score tracking that do not always align with Malaysian financial practices (Chen et al., 2020). Meanwhile, local applications tend to emphasize comprehensive financial analysis without adequately considering the motivational factors that influence sustained engagement among younger users.

Several Malaysian financial applications have struggled to gain traction with young adults. While Touch 'n Go eWallet has achieved success as a payment platform, it lacks the comprehensive budgeting features that appeal to users seeking financial education (Malaysia Digital Economy Corporation, 2023). BigPay offers budgeting tools but requires multiple app integrations that many students find cumbersome. Similarly, GrabPay provides basic expense tracking without the motivational elements needed to maintain user engagement over time.

Traditional banking apps face similar challenges. Research indicates that a significant proportion of young adults aged 18-25 find conventional banking apps overwhelming and discontinue use within the first month (Asian Banking & Finance, 2023). The pattern that emerges from these local solutions is clear: while the technical capabilities exist, user experience design and behavioral engagement strategies fail to meet the specific needs of Malaysian young adults who are transitioning from financial dependence to independence.

Summary of Key Challenges: The primary financial management challenges facing Malaysian young adults include:

- Cognitive biases that prioritize immediate gratification over long-term financial planning
- Overwhelming complexity in existing financial applications that discourages sustained usage
- Cultural mismatch between Western-designed tools and Malaysian financial practices
- Lack of motivational elements to encourage consistent tracking behaviors
- Insufficient educational support for users developing basic financial literacy

These interconnected challenges necessitate a solution specifically designed for Malaysian young adults that combines practical functionality with behavioral design principles.

1.1.2 Target Demographic Analysis

This project targets two distinct but related segments within Malaysia's young adult population. The first consists of university students aged 18-24 who typically manage monthly allowances ranging from RM500 to RM1,500. For many in this group, this represents their first experience handling personal finances independently. Preliminary discussions with university students suggest that while many attempt various financial tracking methods, maintaining consistent usage remains a significant challenge.

The second segment comprises young professionals aged 22-30, typically entry-level workers earning between RM2,000 and RM4,000 monthly. This group needs tools that strike a balance between sophistication and accessibility, particularly for investment tracking and long-term financial planning as they establish their careers and build financial foundations.

Rationale for Target Selection: These specific age groups and income ranges were selected based on several factors. The 18-30 age range represents a critical transition period from financial dependence to independence, where financial habits and attitudes are still forming and malleable. Survey data from preliminary research indicated that this demographic demonstrates both the highest need for financial literacy support and the greatest potential for adopting digital solutions. The income ranges (RM500-1,500 for students and RM2,000-4,000 for young professionals) represent typical Malaysian financial situations for these groups, as documented by the Department of Statistics Malaysia (2024). These ranges are sufficiently narrow to enable focused feature design while broad enough to accommodate variations within each segment. Additionally, these users possess sufficient smartphone proficiency to adopt mobile financial applications but often lack the financial sophistication targeted by professional-grade financial software.

1.1.3 Technological Foundation

The proposed solution leverages modern cross-platform development technologies to ensure broad accessibility and consistent user experience. Flutter, Google's UI framework, enables unified development for both Android and iOS platforms while maintaining native-like performance (Google Developers, 2024). Python Flask provides a robust backend

infrastructure for data management and user authentication, offering the scalability needed for financial applications (Pallets Projects, 2024). MySQL serves as the database management system, ensuring reliable data storage and retrieval for long-term financial tracking.

Appropriateness for Final Year Project: These technology choices are particularly suitable for a Final Year Project for several reasons. Flutter's single codebase approach significantly reduces development time compared to building separate native applications, making cross-platform delivery feasible within academic timeline constraints. The framework's comprehensive documentation and active community support provide accessible learning resources for student developers. Python Flask's minimalist design philosophy allows rapid API development without requiring extensive backend framework expertise, while its modular architecture supports incremental development aligning with Agile methodology. MySQL's widespread adoption ensures abundant educational resources and troubleshooting support. Collectively, these technologies balance professional-grade capabilities with learning accessibility, enabling the delivery of a functional cross-platform application within the 14-week FYP timeline while providing valuable experience with industry-standard development tools.

1.2 Problem Statement

Malaysian young adults currently lack access to culturally appropriate and engaging financial management tools that effectively combine practical functionality with the behavioral design principles necessary for developing sustainable financial habits. Existing solutions often overwhelm users with excessive complexity or fail to provide sufficient motivational elements for consistent usage, potentially limiting financial literacy development and effective money management skills.

The extent of this problem is well-documented. Bank Negara Malaysia's Financial Capability and Inclusion Survey revealed that financial literacy remains inadequate among a substantial portion of young Malaysians, with many having never created a personal budget (Bank Negara Malaysia, 2019). The Credit Counselling and Debt Management Agency (AKPK) has

reported concerning debt-to-income ratios among young Malaysians, indicating widespread financial management challenges (AKPK, 2023).

The Department of Statistics Malaysia has documented that a large proportion of young adults live paycheck-to-paycheck, despite most expressing a desire to improve their financial management skills (Department of Statistics Malaysia, 2024). Additionally, the Securities Commission Malaysia found that young adult participation in investment activities remains low, with lack of knowledge and confidence cited as primary barriers (Securities Commission Malaysia, 2023).

A preliminary survey conducted across several Malaysian universities revealed that most students rely on manual tracking methods — such as notebooks or basic spreadsheets — which they typically abandon within two months. Nearly half of the respondents had attempted multiple financial apps but found them either too complex or culturally inappropriate for their needs.

Core Problem Summary: The fundamental problem is that Malaysian young adults lack financial management tools specifically designed to address their unique combination of challenges: limited financial literacy, cultural expectations around money management, preference for simple interfaces over complex features, need for motivational support to maintain tracking consistency, and transition status between financial dependence and independence. Existing solutions either target Western markets with different cultural contexts or prioritize feature comprehensiveness over user engagement, leaving Malaysian young adults without effective tools to develop essential financial management capabilities during this critical life stage.

1.3 Project Objectives

1.3.1 Primary Objectives

Objective 1: Develop User-Friendly Financial Tracking System

To develop a user-friendly financial tracking system that provides intuitive expense tracking and budget management suitable for varying levels of financial literacy. Success will be measured by:

- Implementation of at least eight core financial management features including expense categorization, budget creation, spending visualization, and monthly summaries
- Achievement of average task completion times under 30 seconds for common operations such as expense entry and budget checking
- Usability testing scores averaging 4.0 or higher on a 5-point scale across ease of use, clarity of information, and navigation efficiency metrics

Objective 2: Implement Gamification and Behavioral Design

To implement gamification and behavioral design elements that encourage positive financial behaviors and sustained application usage. Measurable outcomes include:

- Development of at least 15 achievement badges tied to specific financial milestones
- Implementation of habit streak tracking with visual progress indicators
- Integration of at least five behavioral nudge mechanisms based on established behavioral economics principles
- Achievement of user engagement metrics showing at least 70% of test users returning to the application within 48 hours during beta testing period

Objective 3: Provide Basic Investment Portfolio Management

To provide basic investment portfolio management capabilities enabling users to manually record investment activities, monitor portfolio performance, and track progress against budgeted targets. Success criteria include:

- Implementation of investment transaction recording for at least three major investment types (stocks, funds, bonds)
- Calculation and display of basic portfolio performance metrics including total invested amount, current value, and percentage gain/loss
- Integration of investment goal setting features with progress visualization
- Provision of educational content covering at least five fundamental investment concepts relevant to Malaysian markets

Objective 4: Ensure Cross-Platform Accessibility

To ensure cross-platform accessibility by delivering a functional mobile application compatible with both Android and iOS platforms. Deliverables include:

- Fully functional application versions for Android 8.0+ and iOS 12.0+
- Consistent feature parity across both platforms with no platform-specific functionality limitations
- Responsive user interface adapting appropriately to screen sizes ranging from 4.5 to 6.5 inches
- Successful deployment to at least 20 test devices representing diverse hardware specifications including budget, mid-range, and flagship models

1.3.2 Secondary Objectives

Beyond the primary goals, this project aims to establish a trust-centered interface design that builds user confidence when handling financial data. This involves creating a professional user interface with clear privacy indicators and secure authentication methods. The application will also generate personalized financial insights, offering straightforward advice based on spending patterns, goal-oriented feedback, and educational content tailored to young adult financial challenges.

Comprehensive usability testing with target demographic users will validate the user experience, ensuring that both the interface and features effectively meet user needs. Success will be measured through usability testing sessions with at least 12 participants from the target demographic, achieving average System Usability Scale (SUS) scores of 70 or above, and incorporating feedback from at least 80% of identified usability issues into final design iterations.

1.4 Project Scope

1.4.1 Included Features (In Scope)

The core financial management features include manual expense entry with intelligent categorization, budget creation and monitoring with visual progress indicators, spending pattern analysis through interactive charts and reports, and monthly and weekly financial

summaries. These features form the foundation of the application's financial management functionality.

Investment tracking capabilities include manual investment portfolio tracking, the ability to record investment purchases and monitor performance, investment goal setting with progress visualization, and basic portfolio analytics and reporting. These features must remain accessible to investment beginners while providing adequate functionality for users developing their investment strategies.

The gamification system includes an achievement badge system for reaching financial milestones, habit streak tracking for consistent logging behavior, progress celebrations and motivational notifications, and user level progression based on financial activities. User management and security features encompass secure authentication and registration, data privacy protection and encryption, user profile management and preferences, and local data storage without third-party banking integration.

1.4.2 Excluded Features (Out of Scope)

Banking Integration: Banking integration capabilities are explicitly excluded from this project. The application will not integrate with third-party banking APIs or provide automatic transaction importing. This exclusion is justified by several factors:

- **Regulatory complexity:** Banking API integration requires compliance with Bank Negara Malaysia guidelines that exceed FYP scope and timeline
- **Data privacy concerns:** Automatic bank access introduces significant data security responsibilities beyond student project capabilities
- **Development complexity:** Secure banking integration would require substantial backend infrastructure development that would compromise time available for core functionality
- **Market access barriers:** Most Malaysian banks require commercial partnerships for API access unavailable to academic projects

Real-Time Market Data: Real-time market data features are excluded to focus on manual tracking functionality and avoid dependencies on external APIs. This decision is based on:

- **Cost constraints:** Reliable financial data APIs require subscription fees unsuitable for student projects
- **Technical complexity:** Integrating multiple data sources for Malaysian stocks, funds, and bonds would significantly increase development complexity
- **Scope alignment:** The educational focus on basic investment tracking does not require real-time data, as users can update manually from publicly available sources

Advanced Financial Planning: Advanced financial planning tools such as retirement planning, tax optimization, or loan calculations fall beyond the current project scope. Exclusion rationale includes:

- **Domain expertise requirements:** Accurate financial planning algorithms require actuarial and tax expertise beyond computer science scope
- **Legal liability concerns:** Providing financial planning advice could create regulatory and liability issues inappropriate for academic projects
- **Complexity management:** These features would require extensive testing and validation that would compromise core functionality development

Multi-User Collaboration: Multi-user collaboration features are not included, allowing development efforts to concentrate on optimizing the individual user experience. This exclusion is justified because:

- **Technical complexity:** Implementing secure multi-user data sharing requires sophisticated access control and synchronization mechanisms
- **Security challenges:** Shared financial data introduces additional privacy and security considerations
- **Target demographic needs:** Research indicates young adults in this age group primarily manage individual finances rather than requiring collaborative features

1.4.3 Project Limitations

Several limitations must be acknowledged. The reliance on manual data entry may impact convenience compared to automated solutions. To mitigate this, the application will implement intelligent expense categorization and quick-entry templates that minimize input time. The user interface will feature large, touch-friendly buttons and auto-complete

suggestions based on common Malaysian spending categories such as mamak, Grab rides, and university fees.

Without real-time market data integration, investment tracking remains limited to user-provided information. To address this, the application will provide educational content about major Malaysian investment platforms including Bursa Malaysia, EPF, and ASNB. Where available, public APIs may be leveraged for basic market information display.

The gamification approach may not appeal uniformly across the target demographic, as cultural preferences can vary significantly. To accommodate this diversity, the application will implement customizable gamification intensity levels, allowing users to enable or disable specific motivational features. Cultural consultation with Malaysian advisors will ensure that achievement systems remain respectful and appropriate.

Technical limitations include dependency on user device capabilities and potential performance variations across mobile platforms. Mitigation strategies include implementing progressive web app capabilities as a backup option, ensuring offline functionality for core features, and conducting extensive testing across budget Android devices commonly used by students.

Looking ahead, the application architecture is designed with API integration capabilities that would allow future connection to Malaysian banking APIs (when regulatory frameworks permit) and investment platforms. This design transforms current manual entry limitations into stepping stones toward comprehensive automation in future iterations.

Summary of Key Limitations: The main limitations affecting this project include:

- Manual data entry requirement, reducing convenience but enhancing mindful spending awareness
- Absence of real-time market data, limiting investment tracking to user-provided updates
- Variable gamification appeal across users with different motivational preferences
- Device-dependent performance variations requiring extensive cross-device testing
- Limited scope for advanced financial planning features beyond basic tracking
- Individual user focus without collaborative or family account features

These limitations represent conscious trade-offs that prioritize development feasibility, data privacy, and core functionality over comprehensive feature sets that would be difficult to implement effectively within FYP constraints.

1.5 Project Significance and Contributions

1.5.1 Societal Impact

This application aims to contribute meaningfully to financial literacy improvement among Malaysian young adults, potentially supporting broader economic resilience and reducing financial vulnerability. By providing accessible and culturally appropriate financial management tools, the project addresses a demonstrated need that extends beyond individual benefit to community-level financial capability enhancement.

Link to Problem Statement: The problem statement identified that Malaysian young adults lack access to culturally appropriate, engaging financial management tools that combine practical functionality with behavioral design principles. This project's societal impact directly addresses this problem by providing a solution specifically designed for Malaysian young adults that incorporates local cultural contexts, behavioral psychology principles, and educational content. By improving individual financial management capabilities, the application tackles the documented issues of low financial literacy, poor debt management, and limited investment participation among Malaysian youth, potentially contributing to improved economic stability for this demographic.

The project applies behavioral psychology principles and gamification strategies to demonstrate how technology can encourage positive habit formation in personal finance management. This approach contributes to understanding how digital interventions can effectively address behavioral barriers to financial literacy development (Fogg, 2009).

1.5.2 Academic and Technical Contributions

From an academic perspective, this project contributes to Human-Computer Interaction research by investigating gamification effectiveness in financial applications and examining behavioral design principles for habit formation in mobile environments. The implementation

demonstrates practical application of Flutter framework for financial applications, contributing to mobile development best practices and cross-platform development methodology.

Connection to Problem Statement: The academic contributions directly respond to the problem's core challenge of insufficient motivational elements in existing financial tools. By systematically applying and documenting behavioral design principles within a culturally specific context, this research provides empirical evidence for how gamification and behavioral nudges can address the psychological barriers (present bias, loss aversion) that prevent consistent financial tracking among Malaysian young adults. The technical contributions demonstrate that professional-grade financial applications can be developed efficiently within academic constraints using cross-platform technologies, establishing a replicable methodology for future student projects targeting similar behavioral change objectives.

The user experience design framework established through this project can serve as a reference for future financial education tools, contributing knowledge regarding culturally appropriate technology design. The design methodology and findings provide guidance for developers creating financial applications for similar demographic groups.

1.5.3 Commercial Potential

The application addresses a gap in Malaysian fintech solutions specifically targeted at young adults. While commercial viability cannot be definitively assessed within the Final Year Project timeframe, the technical architecture provides a foundation for potential future development and integration opportunities.

The current Malaysian fintech landscape focuses primarily on payments (Touch 'n Go, GrabPay) or comprehensive banking (Maybank MAE, CIMB Octo) without specifically addressing young adult financial education needs. SmartFinance occupies a unique position by combining behavioral psychology, cultural appropriateness, and educational content specifically for the 18-30 demographic.

Addressing the Identified Gap: This commercial positioning directly responds to the problem statement's identification that existing solutions either overwhelm users with

excessive complexity or fail to provide motivational elements for consistent usage. By focusing specifically on educational needs and engagement strategies for Malaysian young adults — rather than competing on comprehensive features or payment processing — the application addresses an underserved market segment. This focused approach solves the documented problem of high abandonment rates by prioritizing sustainable engagement over feature quantity.

With over 2.8 million Malaysians aged 20-29 (Department of Statistics Malaysia, 2024) and smartphone penetration exceeding 95% in this demographic, the potential market represents a significant opportunity. University enrollment exceeding 500,000 students provides a concentrated initial market for validation and growth.

Future monetization possibilities could include premium features such as advanced analytics and financial coaching, partnerships with Malaysian investment platforms including ASNB and unit trust companies, and educational content sponsorships from financial institutions seeking to engage young adults. Unlike existing solutions, SmartFinance bridges the gap between simple expense tracking and complex financial management by providing culturally appropriate, educationally focused, and behaviorally designed experiences specifically for Malaysian young adults establishing their first independent financial habits.

The scalable design approach ensures that future expansion remains feasible without requiring complete system redesign, supporting potential long-term development beyond the academic project scope.

1.6 Project Team and Organization

1.6.1 Project Team Structure

The project team consists of the student developer, Siow Chuan Sheng from the RSW Programme, who holds primary responsibility for all development activities. These activities include requirements analysis, system design, implementation, testing, and documentation. This role also encompasses coordinating user research and managing usability testing to ensure comprehensive coverage of both technical development and user experience validation.

Project supervision is provided by Ms. Siti Nadiah binti Nain, who offers technical guidance and academic supervision throughout the project lifecycle. The supervisor's role includes monitoring progress, evaluating milestones, and ensuring both project quality and adherence to academic standards.

Progress Monitoring Mechanisms: Project progress is monitored through several structured mechanisms. Weekly supervisor meetings provide regular touchpoints for progress review, challenge discussion, and guidance on technical decisions. These meetings follow a standardized agenda covering completed work, current challenges, and plans for the coming week. Bi-weekly milestone reviews assess deliverable completion against the project timeline, with formal evaluation of sprint outcomes including functional demonstrations and documentation review. Additionally, the developer maintains a digital project log documenting daily progress, technical decisions, challenges encountered, and solutions implemented, which serves as both a progress tracking tool and reference for final project documentation. This multi-layered monitoring approach ensures early identification of delays or challenges, enabling timely intervention and adjustment of project plans when necessary.

1.6.2 Stakeholder Identification

Primary stakeholders include the target users — Malaysian university students and young professionals aged 18-30 — who represent the principal beneficiaries of the project outcomes. Academic supervisors and the evaluation committee constitute critical stakeholders responsible for project assessment and academic validation. The Faculty of Computing and Information Technology at Tunku Abdul Rahman University of Management and Technology serves as the institutional stakeholder, providing resources, guidance, and the academic framework necessary for project completion.

Secondary stakeholders include the university student community participating in testing and feedback provision, contributing to user experience validation and system refinement. The broader Malaysian fintech ecosystem represents an additional stakeholder group with potential interest in project outcomes for future development opportunities.

1.7 Development Methodology

The project employs Agile Development methodology integrated with User-Centered Design principles and iterative usability testing. This approach enables responsive development and consistent incorporation of user feedback throughout the development lifecycle, allowing adaptation to user needs and technical challenges while maintaining progress toward established objectives.

Project Phase Timeline and Activities:

Phase	Duration	Key Activities	Deliverables
Phase 1: Foundation (FYP1)	Weeks 1-2	Requirements gathering, user research, system architecture design	Requirements document, user personas, system design specifications
	Weeks 3-4	Core expense tracking module development and testing	Functional expense tracking module, unit test suite
	Week 5	Budget management module implementation and integration testing	Integrated budget and expense modules, integration test results
Phase 2: Enhancement (FYP2)	Weeks 1-2	Gamification system implementation	Achievement system, habit tracking, progress visualization
	Weeks 3-4	Investment tracking module development	Investment recording and portfolio analytics features
	Weeks 5-6	System integration, performance optimization	Optimized integrated system, performance test results

	Weeks 7-8	Comprehensive testing, user acceptance testing, refinement	Final application, test documentation, user feedback reports
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Table 1.7.1: Project Phase Timeline and Activities for SmartFinance Development

This table illustrates how Agile sprints structure development across Project 1 and Project 2, with each phase building incrementally upon previous work while maintaining focus on delivering functional components that can be tested and refined based on user feedback.

Feedback Gathering Strategy: User feedback is collected primarily through Google Forms surveys distributed to university students and young professionals within accessible networks. The surveys assess current financial management practices, app preferences, and feature priorities. This approach allows for reaching a reasonable number of respondents while remaining feasible within the constraints of a university Final Year Project.

Usability testing is conducted with fellow students and peers who fit the target demographic profile. Participants are asked to use the application and provide feedback on user interface design, feature functionality, and overall user experience. Testing sessions are kept informal and practical, focusing on identifying usability issues and gathering suggestions for improvement.

Testing Approach: Initial testing uses low-fidelity prototypes (wireframes and mockups) shared with classmates and friends to validate basic design concepts and user flow. Feedback at this stage focuses on whether the information architecture makes sense and if navigation feels intuitive.

As development progresses, a working version of the application is shared with volunteer testers from the university community. These testers use the app for a short period and provide feedback through Google Forms or informal discussions. This practical approach ensures that user perspectives inform the design while remaining realistic for a student-led project.

The feedback integration process involves reviewing survey responses and tester comments regularly, identifying common themes or critical issues, and prioritizing improvements based

on feasibility and impact. Changes are implemented iteratively during development sprints to ensure continuous improvement based on actual user input.

1.8 Chapter Summary and Evaluation

This chapter has established the foundation for the SmartFinance project by identifying the financial literacy challenges facing Malaysian young adults and proposing a comprehensive technological solution that combines practical financial management with behavioral design principles necessary for developing sustainable financial habits.

Several key outcomes have been achieved in this chapter. The problem statement has been clearly defined with supporting evidence from preliminary research and published studies. Comprehensive project scope definition, with explicit inclusions, exclusions, and limitations, provides clear boundaries for development activities and stakeholder expectations. Well-structured objectives addressing both technical implementation and user experience goals ensure thorough coverage of project requirements and success criteria.

The project's strengths include its focus on an underserved demographic with demonstrated need, integration of behavioral psychology principles with practical financial tools, appropriate technology selection for cross-platform accessibility, and realistic scope management within Final Year Project timeframe constraints. These strengths provide confidence in project feasibility and the potential for successful outcomes.

However, several risks and challenges must be acknowledged. Adoption challenges may arise if young adults resist another financial application despite stated interest. User research through Google Forms surveys and iterative design based on peer feedback will help mitigate this risk. The technical learning curve for Flutter and Flask development may exceed initial estimates, necessitating allocated buffer time for technical challenges, simplified backup feature implementations, and leveraging university technical support resources.

Data privacy concerns inherent in financial application development require conservative privacy-first design that exceeds Malaysian Personal Data Protection Act requirements, with transparent explanations of data handling in user-friendly language. The sustainability of user engagement presents another challenge, as gamification might create short-term engagement without fostering long-term behavior change. Addressing this requires basing gamification on

behavioral psychology research rather than superficial point systems and validating effectiveness through feedback from student testers.

Areas requiring continued development include gathering user feedback through accessible survey methods, conducting competitive analysis to validate unique value proposition claims, refining technical architecture based on detailed system requirements, and implementing practical testing strategies with available resources. These development areas provide clear direction for subsequent project phases while acknowledging the iterative nature of the development process and the practical constraints of a university Final Year Project.

Connection to Subsequent Chapters: This introductory chapter establishes the foundation that subsequent chapters will build upon systematically. Chapter 2 (Literature Review) will provide the theoretical and empirical evidence supporting the design decisions outlined here, examining financial literacy research, behavioral economics principles, gamification effectiveness studies, and user experience design best practices that inform the proposed solution. Chapter 3 (Methodology and Requirements Analysis) will detail the systematic procedures for translating the problems identified in this chapter into specific technical requirements and design specifications, explaining how user needs will be captured and validated throughout development. Chapter 4 (System Design) will present the technical architecture that addresses the objectives defined here, demonstrating how the identified challenges will be solved through specific design patterns, algorithms, and interface decisions. Together, these chapters form a cohesive narrative showing progression from problem identification through solution design to implementation planning.

This chapter successfully establishes project viability and provides clear direction for subsequent development phases, ensuring a solid foundation for creating a meaningful contribution to financial technology solutions supporting Malaysian young adults' transition from financial dependence to independence.